THE COZY SCIENCE LAB · ORIGINAL STUDY SERIES

Life Science

A Grade 7–10 study handbook with concept notes, examples, self-check explanations and labelled illustrations.

In this handbook

- Cells
- Human Body Systems
- Genetics & Evolution
- Ecosystems & Biodiversity
- Classification
- Photosynthesis & Respiration

Edition 1 · Lab-authored learning materials. Supporting publications are credited with each topic.
Examples are constructed for teaching; diagrams state their model and scale limits. For current Philippine hazard or weather decisions, consult the responsible agencies rather than a static study guide.

Cells

Cells are the basic units of living things. Learning what their parts do helps explain growth, transport and how organisms obtain energy.

Shared foundations

Cells have a cell membrane, cytoplasm, genetic material and ribosomes. Plant and animal cells are eukaryotic: their DNA is mainly inside a nucleus. Bacteria are cells too, but they do not have a membrane-bound nucleus.

Plant and animal cells

Both usually contain mitochondria and carry out cellular respiration. Plant cells have a cellulose cell wall and commonly a large central vacuole. Photosynthetic plant cells contain chloroplasts; not every plant cell has them.

Moving materials

Diffusion is the net movement of particles down a concentration gradient. Osmosis concerns water moving across a selectively permeable membrane. Active transport requires an energy input to move substances against an electrochemical gradient.

Key terms

Organelle

A specialized cell structure with a particular role.

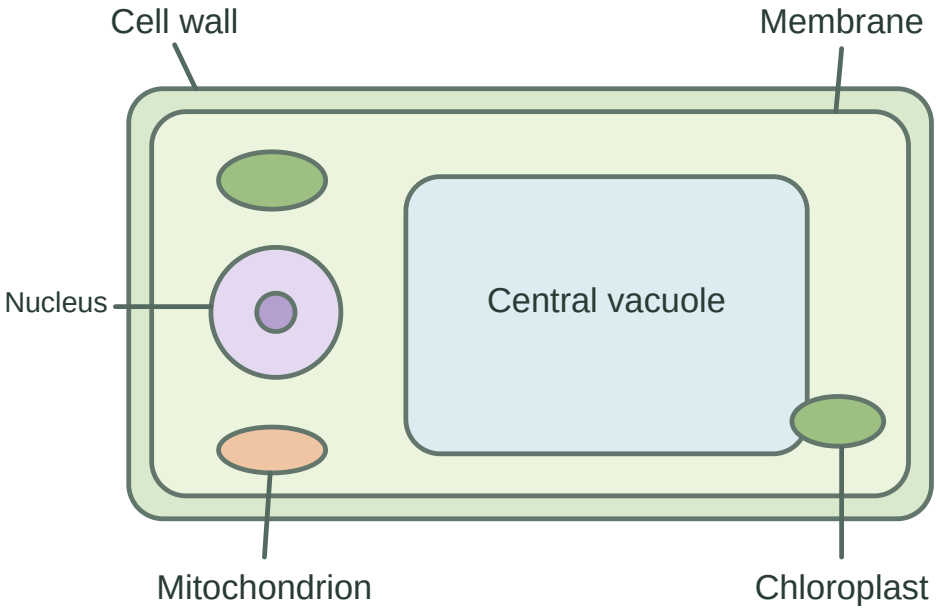
Selectively permeable

Allows some substances to cross more readily than others.

A root cell and a leaf cell

A photosynthetic leaf cell may contain many chloroplasts. A root cell underground normally does not photosynthesize. Both are living plant cells and need cellular respiration.

Remember: A cell wall supports a plant cell; the cell membrane still regulates exchange. A nucleus is not present in every kind of cell.



Inside a plant cell. Schematic, not to scale. Shapes and numbers vary among real cells. Many root cells do not contain chloroplasts.

Check your thinking

Why can an animal cell respire even though it has no chloroplasts?

Explanation: Cellular respiration uses chemical energy from food. Chloroplasts carry out photosynthesis; they are not required for respiration.

Supporting references

These original notes summarize school science concepts. They are not quotations or official curriculum pages.

OpenStax · Rice University. Biology 2e. https://openstax.org/details/books/biology-2e

Philippine Department of Education. K to 12 Science Curriculum Guide. https://www.deped.gov.ph/wp-content/uploads/2019/01/Science-CG_with-tagged-sci-equipment_revised.pdf

Human Body Systems

Body systems cooperate. A useful way to study them is to follow a substance, such as oxygen or a nutrient, from one system to another.

Exchange and transport

The respiratory system exchanges gases with the environment. Oxygen moves from the lungs into blood, and circulation carries it toward tissues. Cells use oxygen in aerobic respiration; breathing and cellular respiration are different processes.

Food and waste

Digestion breaks food into absorbable substances. Most nutrient absorption occurs in the small intestine. Kidneys help regulate water and dissolved substances and remove wastes from blood; this is different from removing undigested food.

Coordination and defence

The nervous and endocrine systems help coordinate responses. Immune defences recognize and respond to threats. Healthy function depends on cooperation among organs, tissues and cells.

Key terms

Tissue

A group of cells working together in a function.

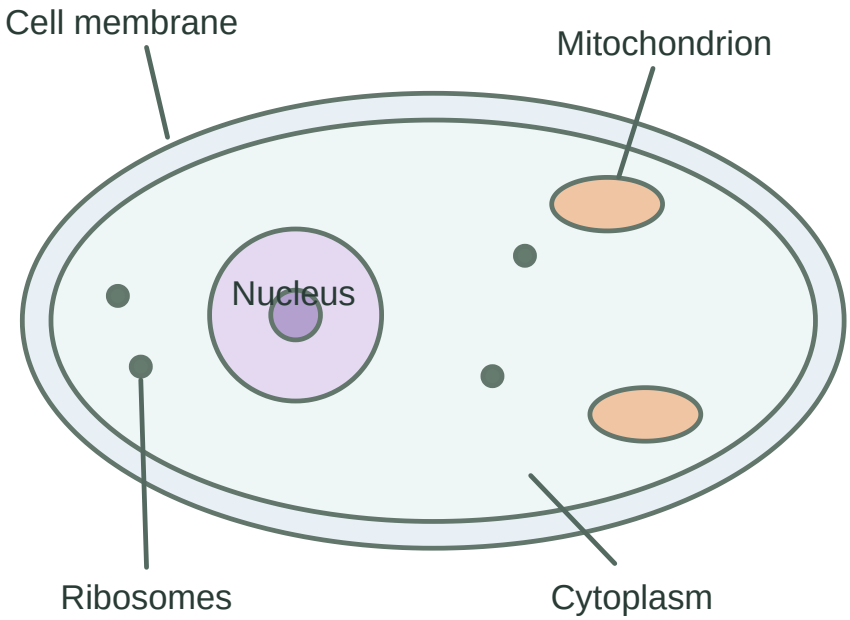
Homeostasis

Regulation that keeps internal conditions within workable ranges.

Following oxygen

Air reaches the lungs, oxygen crosses into blood, and the heart helps move that blood to body tissues. Oxygen then enters cells, where it supports aerobic respiration.

Remember: Deoxygenated human blood is dark red, not blue. Blue in a diagram is a colour key, not the actual colour of the blood.



Inside an animal cell. Schematic, not to scale. Real cells have many more structures and varied shapes.

Check your thinking

Why does faster breathing during exercise involve more than the lungs?

Explanation: Working muscles need more energy; respiration, circulation and coordination work together to support gas exchange and transport.

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Genetics & Evolution

Heredity explains how genetic information passes between generations. Evolution concerns changes in populations over generations.

Genes and alleles

DNA carries genetic information. A gene is a region of DNA associated with a functional product; alleles are versions at a genetic locus. Many traits depend on several genes and environmental conditions.

Simple inheritance models

A Punnett square tracks possible allele combinations under stated assumptions. For a simple dominant/recessive model, genotype describes the allele combination and phenotype describes an observable characteristic.

Natural selection

Heritable variation, differences in survival or reproduction, and repeated generations can change a population. Individuals do not evolve a needed inherited trait simply because they try to adapt.

Key terms

Genotype

The allele combination considered in a genetic model.

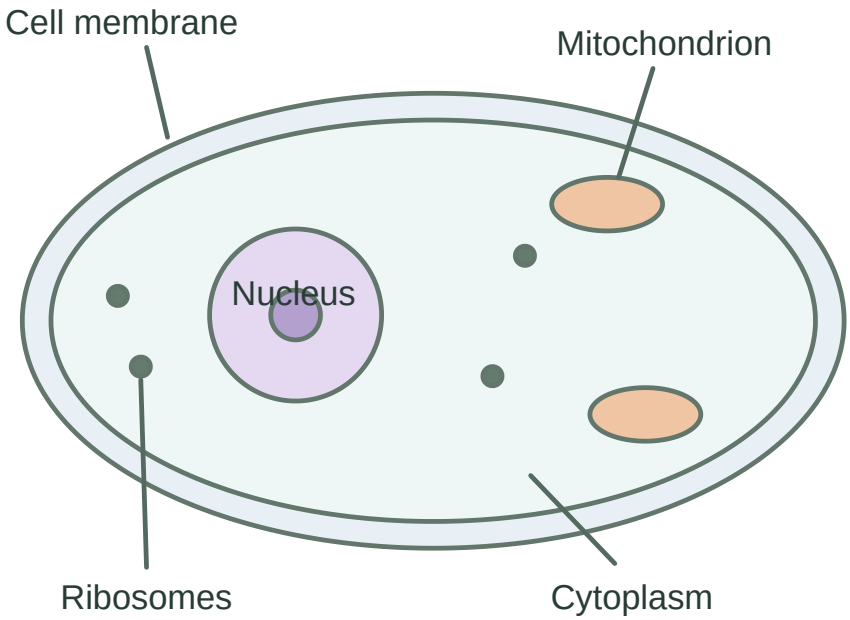
Variation

Differences among individuals in a population.

Aa × Aa

Each parent supplies A or a with equal probability in this simple model. The possible genotypes are AA, Aa, Aa and aa: probabilities 1/4, 1/2 and 1/4. These are probabilities, not a guarantee for four offspring.

Remember: Dominant does not mean more common, better or stronger. Real inheritance is often more complex than a one-gene model.



Inside an animal cell. Schematic, not to scale. Real cells have many more structures and varied shapes.

Check your thinking

Does an individual animal evolve during its lifetime?

Explanation: An individual develops and can respond to its surroundings. Evolution describes changes in populations across generations.

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Ecosystems & Biodiversity

An ecosystem includes organisms and their physical surroundings. Follow both energy and matter to understand how its parts connect.

Energy enters and moves

Producers build organic material using an energy source, often sunlight. Consumers obtain chemical energy by eating. Food-chain arrows point from a food source toward the organism receiving its matter and energy.

Matter cycles

Decomposers break down organic material and help return nutrients to the environment. Matter can cycle through living and nonliving parts. Energy flows through the system and is increasingly dispersed as heat.

Biodiversity and change

Biodiversity includes variation within species, among species and among ecosystems. Changes to habitat or a population can affect feeding relationships and other interactions, not just one isolated species.

Key terms

Population

Members of one species living in a defined area.

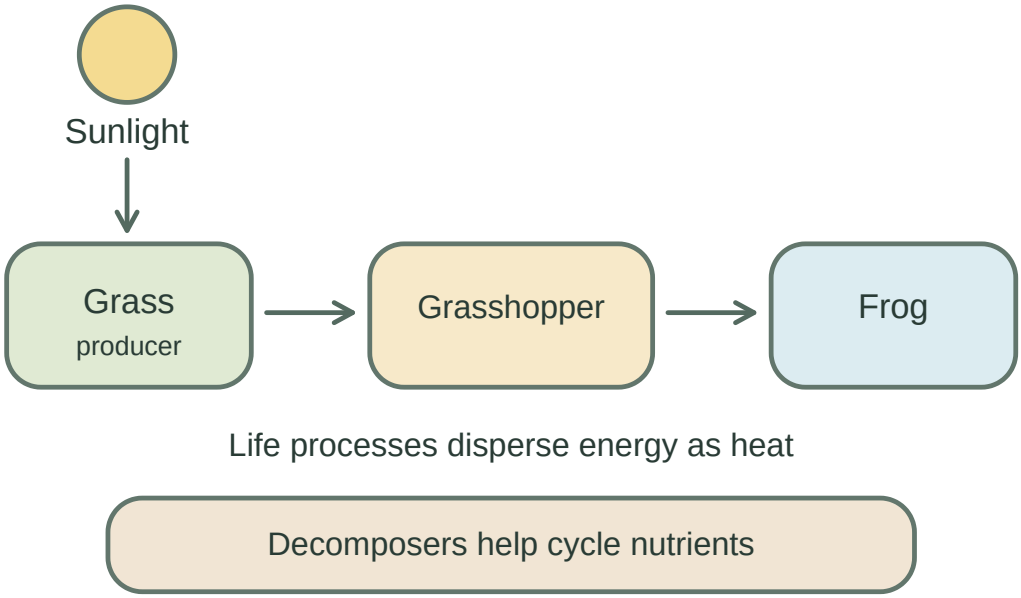
Food web

A model showing interconnected feeding relationships.

Grass → grasshopper → frog

The arrow from grasshopper to frog means the frog obtains material and energy by eating the grasshopper. It does not mean the grasshopper eats the frog.

Remember: Energy-transfer percentages vary. The often-used 10% value is a rough teaching model, not a universal law for every food chain.



Energy flows; matter cycles. A teaching model, not a complete food web. Transfer efficiencies vary; arrow directions identify the receiver of material and energy.

Check your thinking

Do decomposers recycle all the energy in dead organisms?

Explanation: They recycle nutrients and other matter. Energy is transferred and dispersed as heat; it does not cycle back in the same way.

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Classification

Classification helps us communicate about biodiversity and investigate relationships among organisms.

Nested groups

Organisms can be grouped in a hierarchy. Common ranks include domain, kingdom, phylum, class, order, family, genus and species. A species name normally combines the genus and a specific epithet.

Evidence of relationships

Scientists compare structures, development and molecular evidence such as DNA. Similar appearance can be informative, but it can also result from similar ways of living rather than close ancestry.

Models can change

Classification reflects evidence and can be revised. Bacteria and Archaea are distinct domains in a widely used modern framework. School kingdom schemes are useful models, not an unchanging count accepted in every context.

Key terms

Taxonomy

Naming and classifying organisms.

Common ancestor

An ancestral population shared by descendant lineages.

A scientific name

In Homo sapiens, Homo is the genus and sapiens is the specific epithet. The genus begins with a capital letter; the epithet does not. Both words are conventionally italicized when typeset.

Remember: Size, habitat or the ability to fly alone cannot establish a close evolutionary relationship.

Check your thinking

Why might a classification change after DNA evidence is studied?

Explanation: New evidence can support different relationships from those inferred using appearance alone.

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Photosynthesis & Respiration

Photosynthesis and cellular respiration connect energy transformations with the cycling of carbon, oxygen and water.

Capturing light energy

Photosynthetic organisms use light energy to build organic substances from carbon dioxide and water. In plants, photosynthesis occurs in chloroplasts and releases oxygen. Light supplies energy, not the carbon atoms in a plant.

Releasing usable chemical energy

Aerobic cellular respiration transfers chemical energy from food into forms cells can use, including ATP, while releasing carbon dioxide, water and heat. Plant cells respire as well as photosynthesize.

Following the matter

Carbon in newly formed plant material comes largely from carbon dioxide. Water supplies other atoms and helps support growth. Mineral nutrients from soil matter, but soil is not the main source of a plant's dry mass.

Key terms

Chlorophyll

A pigment involved in absorbing light for photosynthesis.

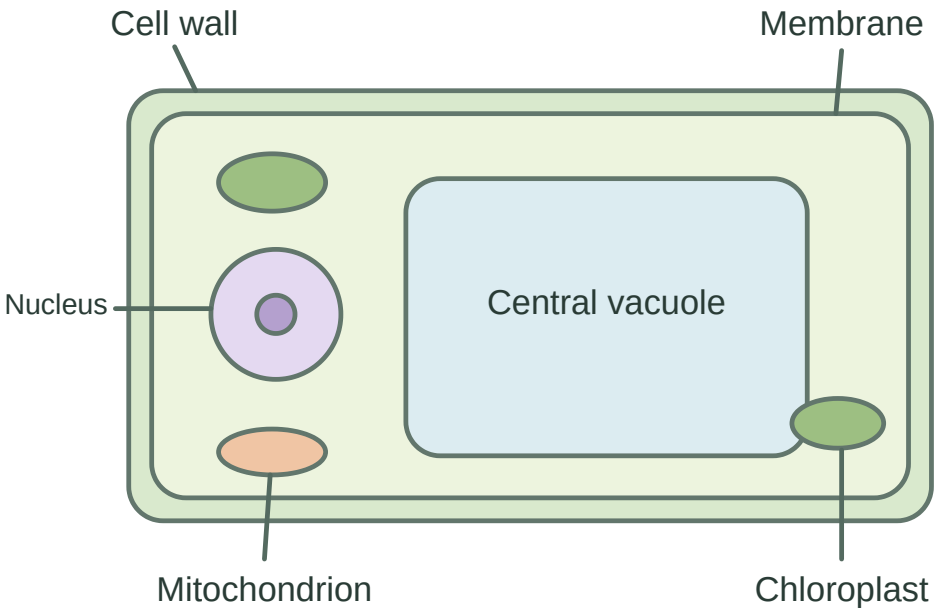
ATP

A molecule used in transferring energy within cells.

A plant at night

In darkness, a typical green plant cannot carry out light-dependent photosynthesis, but its living cells continue cellular respiration using available chemical energy.

Remember: Plants do not only respire at night. Photosynthesis and respiration are not simply the same process running backwards inside the same organelle.



Inside a plant cell. Schematic, not to scale. Shapes and numbers vary among real cells. Many root cells do not contain chloroplasts.

Check your thinking

Where does most of the carbon in a growing tree come from?

Explanation: Carbon dioxide taken from the surroundings supplies the carbon that photosynthesis incorporates into organic molecules.

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